

VM Cost Estimation

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Executive Summary

For a single-VM deployment of ChatProxy Platform in the remote-model scenario, the lowest credible production-like starting point is not the app resource floor from the repo. The app floor is only about **1.75 vCPU / 3.5 GiB RAM**, and that excludes the two MongoDB containers, two PostgreSQL containers, Docker overhead, OS overhead, logging, and headroom.

Using public pricing evidence from AWS and Azure, the practical cost ranges are:

Tier	AWS Base Monthly	Azure Base Monthly	Notes
Budget floor	\$146.62	\$163.57	4 vCPU, 16 GiB RAM, 256 GiB SSD
Safer starting point	\$293.25	\$324.34	8 vCPU, 32 GiB RAM, 512 GiB SSD

My recommendation for a **100 concurrent students** target is to start planning around the safer tier, not the budget floor.

Scope and Assumptions

This estimate assumes:

- single Linux VM
- Docker-based deployment
- one host running the application services plus local databases
- remote-model usage, where Flowise calls external AI APIs
- no local GPU inference
- on-demand pricing only
- region comparison: AWS **us-east-1**, Azure **eastus**

Important clarification:

- this single-VM assumption is a hypothetical consolidation cost model
- it is not the same as the current AWS Terraform deployment in this repo
- the current AWS IaC deploys Flowise and Flowise Proxy as ECS Fargate services, not as one EC2 VM each
- the current Terraform CPU and memory reservations appear to be demo-only sizing, not a production target for **100 concurrent students**
- for the production VM interpretation in this document, the intended shape is **2** Linux VMs with Flowise on its own VM
- for simplicity, that two-VM production interpretation is priced here as **2x** the selected single-VM shape

This estimate excludes:

- outbound bandwidth charges
- public IP charges

- monitoring and logging add-ons
- AI API or token spend in the base VM estimate, covered separately in the AI token addendum below
- backup tooling beyond raw snapshot storage
- labor and operations

Why These VM Sizes Were Chosen

The repo's declared application reservations add up to roughly:

- 1.75 vCPU
- 3.5 GiB RAM

That reservation floor should be treated as a demo footprint only, not as evidence that the full platform can support 100 concurrent students at that size.

That number is too low to use directly for a single production-style VM because it excludes:

- MongoDB for auth
- MongoDB for chat history and sessions
- PostgreSQL for accounting
- PostgreSQL for Flowise
- OS and Docker overhead
- performance headroom for concurrent chat spikes

So the pricing research compared two practical sizes:

Budget floor comparison

- 4 vCPU
- 16 GiB RAM
- 256 GiB SSD

Safer starting point comparison

- 8 vCPU
- 32 GiB RAM
- 512 GiB SSD

AWS Estimate

AWS evidence used

- EC2 on-demand pricing page confirms compute is billed separately from storage and Linux is billed per second.
- EBS pricing page confirms gp3 storage pricing is separate and includes baseline 3,000 IOPS and 125 MB/s.
- Targeted web evidence plus Vantage instance pages gave exact instance-level hourly prices.

AWS instance candidates

Instance	vCPU	RAM	Hourly	Monthly Compute
m6a.xlarge	4	16 GiB	\$0.1728	\$126.14
m6a.2xlarge	8	32 GiB	\$0.3456	\$252.29
m7i.xlarge	4	16 GiB	\$0.2016	\$147.17
m7i.2xlarge	8	32 GiB	\$0.4032	\$294.34

AWS storage assumptions

Using gp3 at **\$0.08 / GB-month**:

Disk Size	Monthly Disk Cost
256 GiB	\$20.48
512 GiB	\$40.96

AWS base totals

Tier	Shape	Base Monthly Total
Budget floor	m6a.xlarge + 256 GiB gp3	\$146.62
Safer starting point	m6a.2xlarge + 512 GiB gp3	\$293.25

AWS interpretation

- **m6a** is the cost-efficient AWS baseline for this use case.
- **m7i** is valid if you want newer Intel hardware, but it raises cost without changing the memory ratio.
- The **m6a.xlarge** tier is a thin floor, not a comfortable 100-concurrent-student target.

Azure Estimate

Azure evidence used

- The official Linux VM pricing page lists **D4as v5** at **\$125.56/month** and **D8as v5** at **\$251.12/month**.
- The Azure retail pricing tool confirmed the same shapes at **0.172/hour** and **0.344/hour** in **eastus**.
- The Azure retail pricing tool provided managed disk pricing for **P15 LRS** and **P20 LRS**.

Azure VM candidates

VM	vCPU	RAM	Hourly	Monthly Compute
D4as v5	4	16 GiB	\$0.172	\$125.56
D8as v5	8	32 GiB	\$0.344	\$251.12

Azure storage assumptions

Disk SKU	Size Class	Monthly Disk Cost
P15 LRS	256 GiB	\$38.01
P20 LRS	512 GiB	\$73.22

Azure base totals

Tier	Shape	Base Monthly Total
Budget floor	D4as v5 + P15 LRS	\$163.57
Safer starting point	D8as v5 + P20 LRS	\$324.34

Azure interpretation

- Azure is slightly more expensive than AWS in this single-VM comparison.
- The **D4as v5** tier is the cheapest credible Azure floor.
- The **D8as v5** tier is the more realistic place to start if the target really is about **100 concurrent students**.

Provider Comparison

Budget floor

Provider	Compute + Disk
AWS m6a.xlarge + 256 GiB gp3	\$146.62
Azure D4as v5 + 256 GiB premium disk	\$163.57

Difference:

- AWS is about **\$16.95/month** cheaper at the budget floor.

Safer starting point

Provider	Compute + Disk
AWS m6a.2xlarge + 512 GiB gp3	\$293.25
Azure D8as v5 + 512 GiB premium disk	\$324.34

Difference:

- AWS is about **\$31.09/month** cheaper at the safer tier.

Recommendation

If the goal is the cheapest acceptable pilot

Use one of these:

- AWS: m6a.xlarge + 256 GiB gp3 at about \$146.62/month
- Azure: D4as v5 + P15 LRS at about \$163.57/month

Risk:

- this is likely too thin for a stable 100 concurrent students target once the four databases, Docker overhead, and peak bursts are factored in

If the goal is a serious single-VM starting point

Use one of these:

- AWS: m6a.2xlarge + 512 GiB gp3 at about \$293.25/month
- Azure: D8as v5 + P20 LRS at about \$324.34/month

This is the tier I would use as the planning baseline for a single-host deployment intended to support about 100 concurrent students.

HKD Addendum For Logged Usage And Images

Using the live rate captured on 2026-05-06 from Xe:

- 1 USD = 7.8352 HKD

Long-session logging model used here

To estimate the impact of logging all student usage and images, this addendum assumes:

- 100 concurrent students
- 1 chatflow per student session
- long conversation session
- remote-model usage, so the VM is not doing local GPU inference
- session payload dominated by stored images, not by text logs

Reasonable storage profiles per session are:

Session Profile	Approximate Logged Data Per Session
text and usage only	2 MB
long session with 5 images	10 MB
long session with 10 images	18 MB
long session with 10 heavier images	24 MB

Storage growth implied by that model

If each of the same 100 students runs one long session per day, that is about 3,000 sessions/month:

Profile	Monthly New Storage
text and usage only	5.9 GB

Profile	Monthly New Storage
long session with 5 images	29.3 GB
long session with 10 images	52.7 GB
long session with 10 heavier images	70.3 GB

If the platform instead sees about **15,000 sessions/month** through repeated daily session waves:

Profile	Monthly New Storage
text and usage only	29.3 GB
long session with 5 images	146.5 GB
long session with 10 images	263.7 GB
long session with 10 heavier images	351.6 GB

Working interpretation:

- if images stay on the VM disk, **512 GiB** is no longer the comfortable default
- **1 TiB** is the safer single-VM disk size for long-conversation retention with image logging
- if images are moved to object storage instead of the VM disk, the earlier **512 GiB** estimate remains more viable

Recommended single-VM cost in HKD

For the image-logging scenario, the safer planning baseline is **8 vCPU / 32 GiB RAM / 1 TiB SSD**.

Provider	Shape	USD / Month	HKD / Month
AWS	m6a.2xlarge + 1 TiB gp3	\$334.21	HK\$2,618.60
Azure	D8as v5 + P30 LRS	\$386.29	HK\$3,026.66

These are the numbers I would use for a first-pass budget when you want to keep full usage logs and images on the same VM.

Higher-headroom option in HKD

If you expect heavier image upload traffic, higher retention on-box, or noticeably more simultaneous processing overhead, move up one compute tier:

Provider	Shape	USD / Month	HKD / Month
AWS	m6a.4xlarge + 1 TiB gp3	\$586.50	HK\$4,595.31
Azure	D16as v5 + P30 LRS	\$637.41	HK\$4,994.23

Two-year total cost in HKD

If you keep the VM running continuously for 24 months at the same on-demand pricing level, the two-year infrastructure totals are:

Plan	Provider	USD / 24 Months	HKD / 24 Months
Recommended single-VM baseline	AWS	\$8,021.04	HK\$62,846.45
Recommended single-VM baseline	Azure	\$9,270.96	HK\$72,639.83
Higher-headroom single VM	AWS	\$14,076.00	HK\$110,288.28
Higher-headroom single VM	Azure	\$15,297.84	HK\$119,861.64

If production means two Linux VMs with a separate Flowise VM

If your production rule is to run 2 Linux VMs, with one dedicated to Flowise and one for the rest of the platform, then the infrastructure numbers above can be approximated as 2x the selected single-VM shape.

Using the same recommended and higher-headroom shapes:

Plan	Provider	USD / Month	HKD / Month	USD / 24 Months	HKD / 24 Months
Recommended production baseline	AWS	\$668.42	HK\$5,237.20	\$16,042.08	HK\$125,692.90
Recommended production baseline	Azure	\$772.58	HK\$6,053.32	\$18,541.92	HK\$145,279.66
Higher-headroom production	AWS	\$1,173.00	HK\$9,190.62	\$28,152.00	HK\$220,576.56
Higher-headroom production	Azure	\$1,274.82	HK\$9,988.46	\$30,595.68	HK\$239,723.28

This is the right way to read the estimate if you want a separate Flowise VM but are still using a simple two-host multiplier rather than sizing the Flowise VM independently.

Approximate infrastructure cost per long session

If 100 students each run one long session per day, that is about 3,000 sessions/month.

Using the recommended 8 vCPU / 32 GiB / 1 TiB plan:

Provider	HKD / Month	Approximate HKD / Session
AWS	HK\$2,618.60	HK\$0.87
Azure	HK\$3,026.66	HK\$1.01

Important:

- this is only the VM infrastructure cost per session
- it does not include model API or token cost

- it does not include outbound bandwidth, monitoring, or object storage if you later move images off the VM

AI Token Cost Addendum

This section estimates text-token consumption and API spend for a single long Flowise session using:

- two AI agents
- MCP tools
- RAG
- long system prompt
- no KV cache

This is a text-token estimate only. It does not include provider-specific image token billing if uploaded images are sent directly into a multimodal model.

Session token model used here

Assumed session shape:

- 12 user turns in one long session
- 2 model calls per turn because the chatflow uses two AI agents
- long system prompt and agent instructions repeated on every call because there is no KV cache
- one RAG injection and one MCP tool result injection per turn

Token budget:

Component	Formula	Input Tokens
system prompt and agent instructions	24 calls * 3,000	72,000
MCP tool schemas and tool instructions	24 calls * 1,500	36,000
current user turn passed into both agents	24 calls * 250	6,000
RAG context injected once per turn	12 * 2,400	28,800
MCP tool result payload injected once per turn	12 * 1,200	14,400
cumulative replayed history with no KV cache	planning estimate	198,000
total input	n/a	355,200

Output budget:

Component	Formula	Output Tokens
combined output from the two agents	12 turns * 1,400	16,800
total output	n/a	16,800

Working planning estimate per long session:

- input: 355,200 tokens

- output: 16,800 tokens
- total metered text tokens: 372,000

Model pricing used

Model option	Input / 1M tokens	Output / 1M tokens	Pricing source used
Azure OpenAI GPT-5 Global	\$1.25	\$10.00	official Azure OpenAI pricing page
AWS Bedrock Amazon Nova 2 Lite	\$0.30	\$2.50	official Bedrock pricing page, live browser-rendered table
xAI Grok 4.3	\$1.25	\$2.50	official xAI API pricing page
OpenRouter Claude Sonnet 4	\$3.00	\$15.00	OpenRouter model page

Estimated token cost per long session

Using 1 USD = 7.8352 HKD:

Model option	USD / Session	HKD / Session
Azure OpenAI GPT-5	\$0.6120	HK\$4.80
AWS Bedrock Amazon Nova 2 Lite	\$0.1486	HK\$1.16
xAI Grok 4.3	\$0.4860	HK\$3.81
OpenRouter Claude Sonnet 4	\$1.3176	HK\$10.32

Estimated token cost for 100 simultaneous long sessions

If 100 students are concurrently in one long session each, the aggregate token volume is roughly:

- input: 35.52M tokens
- output: 1.68M tokens

Estimated API spend for that wave of 100 long sessions:

Model option	USD / 100 sessions	HKD / 100 sessions
Azure OpenAI GPT-5	\$61.20	HK\$479.51
AWS Bedrock Amazon Nova 2 Lite	\$14.86	HK\$116.42
xAI Grok 4.3	\$48.60	HK\$380.79
OpenRouter Claude Sonnet 4	\$131.76	HK\$1,032.36

Estimated monthly token cost at 3,000 long sessions

If 100 students each complete about one long session per day, that is about 3,000 sessions per month.

Model option	USD / Month	HKD / Month
Azure OpenAI GPT-5	\$1,836.00	HK\$14,385.43
AWS Bedrock Amazon Nova 2 Lite	\$445.68	HK\$3,491.99
xAI Grok 4.3	\$1,458.00	HK\$11,423.72
OpenRouter Claude Sonnet 4	\$3,952.80	HK\$30,970.98

Estimated two-year token cost at 3,000 long sessions per month

At the same 3,000 sessions/month level over 24 months:

Model option	USD / 24 Months	HKD / 24 Months
Azure OpenAI GPT-5	\$44,064.00	HK\$345,250.25
AWS Bedrock Amazon Nova 2 Lite	\$10,696.32	HK\$83,807.81
xAI Grok 4.3	\$34,992.00	HK\$274,169.32
OpenRouter Claude Sonnet 4	\$94,867.20	HK\$743,303.49

Combined VM and token total at 3,000 long sessions per month

This combines the recommended single-VM baseline with the token estimate above.

Pairing assumption:

- Azure GPT-5 is paired with the Azure recommended VM baseline.
- AWS Nova 2 Lite is paired with the AWS recommended VM baseline.
- xAI Grok 4.3 and OpenRouter Claude Sonnet 4 are paired with the AWS recommended VM baseline as the lower-cost neutral host.

Model path	USD / Month Total	HKD / Month Total	USD / 24 Months Total	HKD / 24 Months Total
Azure GPT-5 + Azure VM	\$2,222.29	HK\$17,412.09	\$53,334.96	HK\$417,890.08
AWS Nova 2 Lite + AWS VM	\$779.89	HK\$6,110.59	\$18,717.36	HK\$146,654.26
xAI Grok 4.3 + AWS VM	\$1,792.21	HK\$14,042.32	\$43,013.04	HK\$337,015.77
OpenRouter Claude Sonnet 4 + AWS VM	\$4,287.01	HK\$33,589.58	\$102,888.24	HK\$806,149.94

Combined production total with a separate Flowise VM

If production means 2 Linux VMs with one dedicated to Flowise, while token demand stays the same, the combined totals become:

Model path	USD / Month Total	HKD / Month Total	USD / 24 Months Total	HKD / 24 Months Total
Azure GPT-5 + Azure production infra	\$2,608.58	HK\$20,438.75	\$62,605.92	HK\$490,529.91
AWS Nova 2 Lite + AWS production infra	\$1,114.10	HK\$8,729.19	\$26,738.40	HK\$209,500.71
xAI Grok 4.3 + AWS production infra	\$2,126.42	HK\$16,660.92	\$51,034.08	HK\$399,862.22
OpenRouter Claude Sonnet 4 + AWS production infra	\$4,621.22	HK\$36,207.18	\$110,909.28	HK\$869,996.39

If you prefer to host the Grok or OpenRouter path on Azure instead of AWS, add:

- **HK\$408.06 / month**
- **HK\$9,793.38 / 24 months**

Interpretation

- With no KV cache, replayed history dominates input token cost.
- For this session shape, token spend is likely to exceed VM spend for GPT-5, Grok, and Claude.
- Nova is much cheaper in this estimate, but the verifiable AWS Bedrock row available in research was **Amazon Nova 2 Lite**, which is not a like-for-like frontier-model comparison with GPT-5 or Claude Sonnet 4.
- If you later enable prompt caching, shorten the system prompt, reduce RAG chunk size, or collapse the two-agent pattern into one model call on some turns, token spend can fall materially.

Caveats

Local inference is not included

If the same VM must run a local LLM instead of calling remote model APIs, these numbers stop being relevant. That becomes a GPU sizing problem.

Bandwidth is excluded

- AWS includes the first **100 GB** of internet egress per month across services, but higher usage will add cost.
- Azure bandwidth was not priced in this pass because traffic volume is unknown.

Single-VM architecture risk remains

Even at **8 vCPU / 32 GiB**, a single VM is still one failure domain. If the platform becomes important enough that downtime is costly, the next step should be splitting the databases off-box or moving toward managed services rather than only scaling up the VM.

Research Trail

Supporting notes:

- [docs/research/vm_cost_research_plan.md](#)
- [docs/research/aws_vm_pricing_evidence.md](#)
- [docs/research/azure_vm_pricing_evidence.md](#)
- [docs/research/ai_token_pricing_evidence.md](#)